EXP NO: 2

DATE: 23/7/26 UNDERSTAND DIFFERENT TYPES OF NETWORK CABLES

AIM:

Study of different types of Network cables

TYPES OF NETWORK CABLES:

1. Unshielded Twisted pair (UTP) cable
2. Shielded Twisted pair (STP) cable
3. Coaxial cable
4. Fibre optic cable.

CABLE TYPE	CATEGORY	MAXIMUM DATA TRANSMISSION	ADVANTAGES/ DISADVANTAGES	APPLICATIONS/ USE
UTP	Category 3	10 bps	Advantages: • Cheaper in cost • Easy to install as they have a smaller overall diameter	10 Base-T- Ethernet
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast, Gigabit Ethernet.
STP	Category 6, 6a	10Gbps	Advantages: • Shielded • Faster than UTP • Less susceptible to noise & interference Disadvantages: • Expensive • Greater Installation effort	Gigabit Ethernet, 10G Ethernet (55/m Widely used in data centers
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100 m).
coaxial cable	RG-6 RG-59 RG-11	10 – 100 Mbps	• High bandwidth • Low loss bandwidth • Versatile Disadvantages: • Limited distance • Cost • Size is bulky	Speed of signal is soon Television network high speed internet connections
Fibre optics cable	Single mode Multi mode	100 Gbps	• High speed • High security • Long distance Disadvantages: • Expensive • Required skilled	Maximum distance of fibre optics cable is 100 meters.

RESULT:

Thus the study of different types of network cables has been verified.